

Week 2: SOC Operations – Alert to Response Cycle

This report documents theoretical knowledge and practical SOC operations covering alert prioritization, incident classification, response, evidence preservation, and a capstone simulation.

1. Alert Priority Levels

Alert priority defines how urgently an alert must be handled.

Critical alerts involve active exploitation, ransomware, or severe business impact.

High alerts include unauthorized access and privilege escalation.

Medium alerts include brute-force attempts or blocked malware.

Low alerts include reconnaissance and policy violations.

CVSS scoring helps quantify risk. Example: Log4Shell CVSS 9.8 is classified as Critical.

2. Incident Classification

Incident classification ensures consistency and faster investigations.

Common categories include malware, phishing, DDoS, insider threats, and data exfiltration.

MITRE ATT&CK; mapping example: Phishing emails map to technique T1566.

Incidents are enriched with metadata such as affected systems, IP addresses, and timestamps.

3. Incident Response Lifecycle

Preparation includes tools, playbooks, and access control.

Identification involves alert validation and triage.

Containment isolates affected systems.

Eradication removes malicious artifacts.

Recovery restores normal operations.

Lessons learned improve future response.

4. Alert Management Practice

Alerts were categorized and prioritized using Google Sheets.

Mock alerts such as Log4Shell and Port Scans were evaluated.

Incident tickets were created in TheHive.

Escalation procedures were practiced.

5. Response Documentation

Incident response templates were created based on SANS standards.

Investigation timelines and action logs were documented.

Phishing response checklists were prepared.

Post-mortem analysis was conducted.

6. Alert Triage Practice

Wazuh alerts were analyzed for severity and validity.

Threat intelligence validation was performed using VirusTotal and AlienVault OTX.

False positives were identified and documented.

7. Evidence Preservation

Volatile data was collected using Velociraptor.

Memory dumps were acquired and hashed using SHA256.

Chain-of-custody documentation was maintained.

8. Capstone Project

A Metasploit attack was simulated against Metasploitable2.

Wazuh detected and generated alerts.

CrowdSec blocked the attacker IP.

A final incident report and management briefing were created.

Conclusion

This task demonstrates end-to-end SOC operations following industry best practices.